

ECO 202 — In-Class Exam 1

Worked solutions

Fall 2026 | Classes 1–5

Equivalent exact expressions and valid alternative reasoning are acceptable. For practice study, attempt the paper first, then compare reasoning, close these solutions, and reconstruct any missed method independently.

1–4. Multiple-choice answers and reasoning

1. **B.** The record refers to a particular shop in a particular month. The industry numbers label categories, so arithmetic differences between those codes are not industry amounts. A and C misclassify the code and unit; D identifies neither the record nor the variable correctly.
2. **C.** The transformed mean is $10 - 2\bar{x}$ and standard deviation is $2s_x$, so $z_y = [10 - 2x - (10 - 2\bar{x})]/(2s_x) = -z_x = -3/2$. A misses the reflection; B treats a standardized value as an original-scale value; D fails to rescale the standard deviation.
3. **B.** With positive standard deviations, $b_1 = rs_y/s_x = 0$. Zero correlation rules out neither a nonlinear relationship nor a causal mechanism, so A and C fail. The response can vary around its mean, so D also fails.
4. **A.** Time ordering, a large difference, and computational replication are compatible with a confounded observational association. None identifies the missing counterfactual comparison on its own.

5. Describing a distribution

- (a) The median is $(5 + 7)/2 = 6$, $Q_1 = 4$, $Q_3 = 8$, and IQR is 4. A correct plot has a box from 4 to 8, a median line at 6, and whiskers at 2 and 13, on a labeled numerical axis. No point is separately flagged.
- (b) The fences are $4 - (3/2)4 = -2$ and $8 + (3/2)4 = 14$. Since $13 < 14$, it is not flagged. A visibly large value is not automatically an outlier by this particular rule.
- (c) The other five values sum to 26. Set $26 + v = 6(20)$, giving $v = 94$. The new values are 2, 4, 5, 7, 8, 94. Their median is still 6, lower-half median 4, and upper-half median 8, so the IQR remains $8 - 4 = 4$. The mean is $120/6 = 20$. This illustrates the resistance of the median and IQR and the mean's sensitivity to an extreme observation.

6. Units and standardization

- (a) The z-score is

$$z = \frac{30 - 24}{4} = \frac{3}{2}.$$

This worker's hourly wage is 1.5 sample standard deviations above the sample mean. The z-score is unitless.

- (b) Reverse the standardization:

$$x = \bar{x} + zs_x = 24 + \left(-\frac{1}{2}\right)4 = 22 \text{ dollars per hour.}$$

This wage is half a sample standard deviation below the sample mean.

- (c) Writing $y_i = 100x_i$, the new mean is $\bar{y} = 100(24) = 2400$ cents per hour and the new standard deviation is $s_y = 100(4) = 400$ cents per hour. Multiplication by a positive constant multiplies both the mean and the standard deviation by that constant. The variance, rather than the standard deviation, scales by its square.

- (d) For $z_i = (x_i - 24)/4$, the sample mean and sample variance are

$$\bar{z} = \frac{24 - 24}{4} = 0, \quad s_z^2 = \left(\frac{1}{4}\right)^2 s_x^2 = \frac{16}{16} = 1.$$

Subtracting the mean centers the data without changing their spread; dividing by the positive standard deviation makes their sample variance one. Shifting and positive rescaling preserve the shape and order of the distribution, so a right-skewed distribution remains right-skewed. A mean of zero and variance of one do not imply symmetry or Normality. These transformation rules suffice; a summation proof is not required.

7. A descriptive regression

- (a) $b_1 = 6/10 = 3/5$ and $b_0 = 4 - (3/5)3 = 11/5$, so $\hat{y} = 11/5 + (3/5)x$. In this fitted relationship, another hundred square feet is associated with $3/5$ thousand dollars more weekly sales. This is a descriptive association among the five stores.

- (b) At $x = 4$, $\hat{y} = 23/5$. The residual is observed minus fitted, $4 - 23/5 = -3/5$ thousand dollars. This store's observed sales are below the fitted value by $3/5$ thousand dollars.

- (c)

$$r = \frac{6}{\sqrt{10(6)}} = \frac{3}{\sqrt{15}}, \quad r^2 = \frac{36}{60} = \frac{3}{5}.$$

With an intercept, the fitted line accounts for $3/5$, or 60%, of the total squared variation in weekly sales about the sample mean across these five stores. This is a decomposition of sample variation, not a share of sales caused by area and not a guarantee of future prediction accuracy.

- (d) The observed area range is 1 to 5, making $x = 20$ substantial extrapolation. The same straight-line pattern has not been demonstrated there. A causal interpretation separately requires a defensible comparison or design; store location, clientele, or other pre-existing differences could explain the observational association. Fixing the range problem alone would not resolve causality.

8. Treatment effects and counterfactuals

Worker	Observed Y_i	$Y_i(1) - Y_i(0)$
A	4	2
B	9	3
C	3	4
D	7	1

- (a) The observed outcomes and individual effects are listed above. Treated workers reveal $Y_i(1)$, untreated workers reveal $Y_i(0)$.
- (b) The ATE across all four workers is $(2 + 3 + 4 + 1)/4 = 5/2$. The observed difference is $(4 + 9)/2 - (3 + 7)/2 = 13/2 - 5 = 3/2$, in thousands of dollars.
- (c) The missing potential outcomes are $Y_A(0), Y_B(0), Y_C(1), Y_D(1)$. No: a worker's individual treatment effect is $Y_i(1) - Y_i(0)$, which requires both potential outcomes for that same worker. Real data reveal only the outcome under the treatment status that actually occurred. Without the other potential outcome, the observed outcome alone does not determine the individual effect.